

Music Corner

A Feasibility Study for Creating a Gesture-Based Rhythm Game for Music Education Inspired by Solfege Hand Signs

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ABSTRACT

In Hong Kong, the gap in music education access between children from affluent and grassroots families is pronounced. To address this, we present "Music Corner," a prototype of a gesture-based rhythm game designed to democratise music education through digital innovation. Utilising mobile technology and machine learning, this study explores the feasibility of an engaging, accessible, and cost-effective platform for music learning. It leverages the Solfege Hand Signs, used by Hong Kong's primary music education, by transforming them into a game where players interact with musical notes through intuitive gestures. This initiative aims to bridge the educational divide, fostering an inclusive environment where children of all socioeconomic statuses can nurture their musical abilities. The prototype features three game modes to cater different learning objectives. With a focus on interactive technology, "Music Corner" stands as a testament to the potential of gamification in enhancing music literacy, making education more inclusive and enjoyable.

CCS CONCEPTS

• **Applied computing** → Education; E-learning; • **Software and its engineering** → Software organization and properties; Contextual software domains; Virtual worlds software; Interactive games; • **Computing methodologies** → Machine learning; Learning paradigms; Supervised learning; Supervised learning by classification.

KEYWORDS

Music Education, Solfege Hand Signs, Rhythm Game, Hand Pose Recognition, Mobile Application

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1 BACKGROUND AND MOTIVATION

In Hong Kong, despite its wealth, poverty limits many children's potential, with the HKSAR encouraged 'one student one instrument' scheme [1] adding financial strain on grass-roots families who struggle to afford this educational enrichment. While twelve years of free education are provided, the shift towards a holistic curriculum introduces extra costs beyond the reach of the underprivileged, perpetuating inequality. Christian Action works with government initiatives to support these families, offering educational and skill development opportunities to help children overcome economic hurdles and realise their dreams [2]. However, such support covers only a very limited number of grassroots families.

The pandemic's push towards blended learning, supported by government-funded smart devices, opens doors for musical education apps like the Education University of Hong Kong's e-Orch [3]. Our objective is to develop accessible, interactive music apps that make learning inclusive for all children, using Solfege Hand Sign-inspired games to democratize music education.

1.1 Solfege Hand Sign and Music Education

Solfege hand signs, dating back to the 11th century and refined by the Kodály Method, play a crucial role in music literacy [4]. Embedded in Hong Kong's music education [5], these signs aid memory and enable a multisensory approach to learning pitch. They simplify note recognition by replacing letters C to B with solfege syllables—do, re, mi, fa, sol, la, ti—making sight-reading more instinctive [6]. This not only helps students quickly identify musical key centers but also improves singing accuracy, a valuable skill for learners with or without musical backgrounds. The tradition of solfege connects generations and democratizes music education, maintaining its relevance and appeal.

1.2 Integrating Solfege Hand Signs into a Rhythm Game

Illustrated in Figure 1, there are games inspired by Solfege Hand Signs, like the basic "Solfege Matching" which lacks musical integration, or "Solfa!" and "Solfa Land" that necessitate physical printing for play, seem more oriented towards memorization of hand signs through gameplay rather than facilitating actual music learning.

To elevate the Music Garden concept and distinguish it from existing practices, our team developed a rhythm game prototype that transforms Solfege Hand Signs into instruments to diversify music education. We integrate hand-tracking technology into Solfege



Figure 1: Games based on Solfege Hand Sign – “Solfege Matching”, “Solfa!” and “Solfa Land”

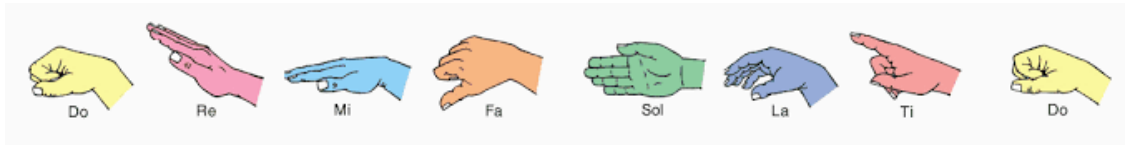


Figure 2: Solfege Hand Signs, via Music Theory Tutor. (<https://www.musictheorytutor.org/wp-content/uploads/2012/10/kodalyhand-rows.png>)

Hand Signs. Players emulate solfege gestures illustrated in Figure 2, guided by avatars, to match on-screen notes, receiving instant feedback on their performance, blending learning with rhythm gaming excitement. This game will use smart devices and gamification to make learning music more accessible and fun.

Unlike static games, these features would make the Music Corner a dynamic space where music theory is not just learned but experienced through play. This gameplay mechanic would turn the player’s hand movements into a musical instrument. Each correct hand sign would not only score points but also play the associated note, allowing the player to effectively ‘play’ the song through their gestures. This immersive experience would not only reinforce the learning of solfege but also provide a sense of accomplishment as players hear their actions contributing to the music.

2 RESEARCH QUESTION AND DEVELOPMENT METHODOLOGY

The Research Question of this study is “How can the machine learning model and mobile technology to support the development of a gesture-based rhythm game utilising Solfege Hand for bridge the educational divide in music literacy for children varying socioeconomic statuses in Hong Kong?”

The study used the Scrum framework for mobile application design and development. It emphasized feedback from educators and students, as well as insights from competitions. Scrum is an agile project management approach that emphasizes iterative and incremental development, collaboration, and adaptability [7]. It enables rapid prototyping and continuous improvement, making it suitable for developing innovative educational tools [8].

3 PROTOTYPE GAME DESIGN AND FEATURES

“Music Corner” is a prototype game designed to blend the Kodaly method with hand tracking technology, enabling children to learn musical scales through solfege hand signs. It’s a virtual space that

combines visual, auditory, and kinesthetic learning to enhance traditional music education [9]. The game offers three modes, each focusing on different educational aspects. The game play is presented in Figure 3.

Recognition Mode: The recognition Mode was tailored to enhance the player’s ability to recognise different musical scales, this mode randomly generates scales for users to identify. A time limit adds a gamified challenge, encouraging players to quickly learn and differentiate scale gestures and their corresponding sounds.

Free Play Mode: Free play mode allows children to select music of their choice, this mode displays the song’s scales on-screen, enabling users to play the music using gestures. This mode is based on the premise that the freedom to interact with music will spark creativity and enjoyment.

Rhythm Mode: Rhythm mode was designed as the most challenging aspect of the app, this mode invites children to play songs of varying difficulty levels. As the difficulty increases, the tempo of the scale cues escalates, testing the player’s timing and accuracy in gesture performance. Scores are based on the consistency and correctness of the player’s gestures, with an optional prompt toggle available for assistance.

4 PROTOTYPE GAME DESIGN AND FEATURES

Our “Music Corner” app focuses on two main objectives: harmonising with iPads used in Hong Kong’s education and developing a robust solfege hand pose recognition algorithm. We chose iOS for its widespread use in schools, ensuring our app aligns with existing technology. We’ve employed Apple’s Vision framework for its precision in translating hand movements into feature points [10], central to our gesture detection. Additionally, we use Apple’s CoreML and CreateML for training our classifier, allowing us to efficiently teach our system to identify solfege signs without complex coding, thanks to CreateML’s user-friendly interface [11].

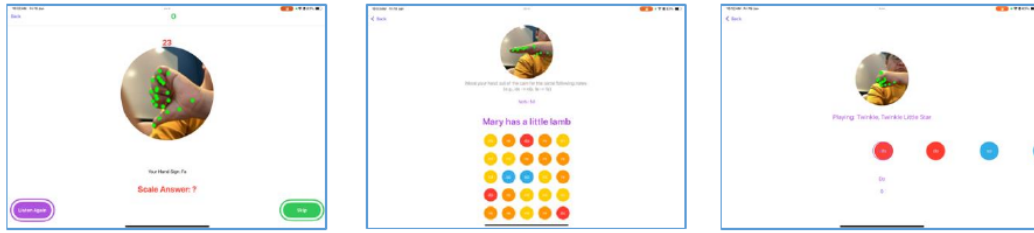


Figure 3: Game Play of Music Corner in Recognition Mode, Free Play Mode and Rhythm Mode.



Figure 4: The 21 feature points detected by Apple's Vision framework.

4.1 Training Machine Learning Model

The mechanism for recognising hand poses within Music Corner relies on a process that combines data collection, hand pose estimation, and machine learning classification:

Data Collection: An extensive dataset of hand images for each Solfege tone ('do' to 'ti') is compiled, with 100 samples per tone captured under various conditions, categorised for training, validation, and testing.

Hand Pose Estimation: Utilising the Vision framework within CreateML, the system maps 21 feature points on the hand illustrated by Figure 4, translating poses into numerical representations based on anatomical landmarks.

Machine Learning Classification: Feature points are fed into Create ML's classifier to discern patterns corresponding to musical tones. The model predicts the probability of each tone, including the chance of 'nothing', indicating no hand pose.

4.2 Integrating the Machine Learning Model into iOS App

The CreateML Tool is then export the pre-trained model in mlmodel format. The mlmodel can be imported into iOS development project

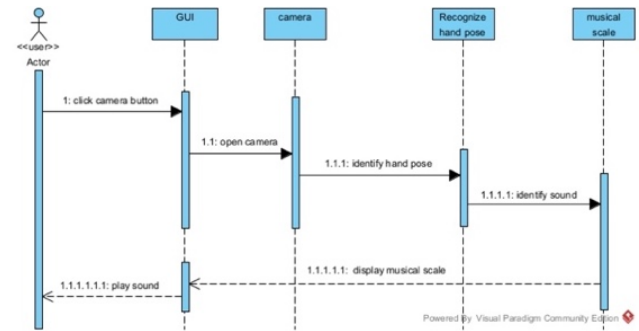


Figure 5: Sequence diagram of real-time video recognition of Music Corner.

directly for recognition. The system processes are presented in Figure 5. The game captures live video from an iOS device's camera to detect the user's hand movements. Using the Vision Framework, it analyses each frame for hand pose estimation, identifying 21 feature points. These points are instantly fed into a pre-trained AI model, which has learned to associate hand poses with musical tones. The model classifies the hand pose and determines the corresponding musical tone. Upon classification, the system plays back the tone, providing real-time auditory feedback.

5 FINAL OUTCOME AND FINDINGS

The Music Corner app's latest update features a machine learning model with 90% accuracy in hand sign recognition, enhancing user experience through seamless interaction. Remarkable for its low-latency on iPads, the app provides instant feedback, crucial for real-time learning. With stable performance across three game modes, the app demonstrates flexibility and developer dedication, establishing it as a reliable, adaptable, and fun tool for music education.

6 DISCUSSION AND LIMITATIONS

During the course of user testing, with students from Chan Creative School as initial testers, the feedback received was largely positive, suggesting that the app has the potential for a positive reception on a broader scale. Despite this, there are several identified limitations that warrant attention. The app currently struggles with differentiating between multiple users, which is a critical feature

for group interactions and multiplayer functionalities. This limitation impedes the app's potential in settings such as classrooms where collaborative learning is key. Furthermore, the system's performance is significantly diminished under low-light conditions, and there are difficulties in recognising repeated notes and distinguishing between octaves due to the limitations in the app's current recognition algorithms.

7 CRITICAL REFLECTION

The game "Music Corner" addresses the significant challenge of bridging the gap in music education access between children from affluent and grassroots families in Hong Kong. It aims to democratize music education by leveraging digital innovation to create an engaging, accessible, and cost-effective platform for learning music. Utilizing mobile technology and machine learning, the game transforms Solfege Hand Signs, a component of Hong Kong's primary music education, into an interactive and enjoyable learning experience. Through this gesture-based rhythm game, the initiative seeks to foster an inclusive environment where children of all socioeconomic statuses can develop their musical abilities and literacy.

The potential to help is particularly strong among underprivileged children who may not have access to traditional music education due to financial constraints. The widespread availability of smart devices, bolstered by government-subsidized programs during the COVID-19 period, makes the game a viable option for these children. By integrating education with the excitement of rhythm gaming, the game can make learning music more accessible, enjoyable, and interactive, potentially sparking a lasting interest in music and enhancing cognitive development.

However, there could be a potential to unintentionally harm populations that do not have access to the required technology or the proficiency to use it. The game relies on mobile devices and a certain level of digital literacy, which may exclude children and families who lack these resources or skills. Additionally, the game's reliance on hand tracking and gesture recognition technology may present challenges for children with physical disabilities affecting hand movement, potentially creating a barrier rather than a bridge

in those cases. It's important to note that these limitations can be addressed in future iterations of the game, with the aim of making it more inclusive and accessible to all populations.

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